Module 1: Uncertainty

#	Question	Marks	Month & Year
#	From below given probability distribution find P (Cavity | Toothache)	5	Dec 2023
#	Explain Bayes theorem	5	Dec 2024
#	How does Bayes Theorem determine the probability of an event with uncertain knowledge? Give one example of it	5	Dec 2025
#	Write short note on: Bayes Theorem	5	May 2023
#	A medical test for a disease is 99% accurate. The disease is rare and affects 1 in 10,000 people. Use Bayes' Theorem to calculate the probability that a person who tests positive actually has the disease. Interpret your result	5	Dec 2025
#	Discuss the limitations of deterministic approaches and how probabilistic reasoning addresses these limitations with example	5	Dec 2025
#	Discuss the limitations of deterministic approaches and how probabilistic reasoning addresses these limitations with example	5	May 2025
#	Describe the role of utility and expected utility in decision theory with example	5	Dec 2025
#	Describe the role of utility and expected utility in decision theory with example	5	May 2025
#	Explain Markov Decision Process Model (MDP). List two applications of it	5	Dec 2025
#	What is need of Markov Decision Processes? Explain MDP in detail	10	May 2023
#	What is a Markov Decision Process (MDP)? List its primary components. Explain in detail	10	May 2024
#	Illustrate inferencing in Bayesian Belief Network with an example	10	Dec 2023
#	What is the Bayesian Belief Network? Illustrate with an example	10	Dec 2024
#	What is Bayesian Belief Network? Explain inferencing with example	10	May 2025

Module 2: Cognitive Computing

#	Question	Marks	Month & Year
#	Illustrate usage of taxonomies and ontologies for knowledge representation in cognitive system	10	Dec 2023
#	Explain the role of representing knowledge in taxonomies and ontologies, and how advanced analytics can be applied to cognitive systems	10	Dec 2025
#	Explain the role of representing knowledge in taxonomies and ontologies, and how advanced analytics can be applied to cognitive systems	10	May 2025
#	List and explain the design principles of Cognitive System	10	Dec 2023

#	Question	Marks	Month & Year
#	Write short note on: Design Principles for Cognitive Systems	5	May 2023
#	List Design Principles of Cognitive system and justify the role of NLP in enabling adaptability, contextual understanding, and user interaction within cognitive architectures	10	Dec 2025
#	Define Cognitive Computing. Draw a neat diagram of components of the cognitive system and explain the components	10	Dec 2024
#	What is cognitive computing, and how does it differ from traditional computing? List the process of building a cognitive application	10	May 2024
#	Explain the process of building a cognitive application	10	May 2023
#	Describe Natural Language Processing in Support of a Cognitive System	10	Dec 2024

Module 3: Fuzzy Logic & Its Applications

#	Question	Marks	Month & Year
#	Define defuzzification and state the necessity of the defuzzification process	5	Dec 2023
#	Define Defuzzification. Discuss any two methods of defuzzification	10	Dec 2024
#	Write short note on: Defuzzification Methods	5	May 2023
#	Define Defuzzification? Explain any two methods of Defuzzification	10	May 2025
#	Consider two fuzzy sets. Find the algebraic sum, algebraic product, bounded sum, and bounded difference of the given fuzzy sets	5	Dec 2024
#	Describe the Properties of Fuzzy Sets with an example	10	Dec 2023
#	Explain different types of membership functions used in fuzzy logic with suitable examples. Differentiate between crisp sets and fuzzy sets in terms of their structure, interpretation, and applicability in real-world scenarios	10	Dec 2025
#	Explain different types of membership functions used in fuzzy logic with suitable examples. Differentiate between crisp sets and fuzzy sets in terms of their structure, interpretation, and applicability in real-world scenarios	10	May 2025
#	Differentiate between Fuzziness and Probability	5	May 2025
#	Using Mamdani fuzzy model design a fuzzy model controller to determine the wash time of domestic washing machine. Assume the inputs are dirt & grease on cloths. Use three descriptors for each input variable & five descriptors for output variables. Derive necessary membership functions & required fuzzy rules for the application	10	Dec 2025

#	Question	Marks	Month & Year
#	Using Mamdani fuzzy model design a fuzzy logic controller to determine the wash time of domestic washing machine. Assume the inputs are dirt and grease on clothes. Use three descriptors for each input variable and five descriptors for output variables. Derive necessary membership functions and required fuzzy rules for the application	10	May 2023
#	Using Mamdani fuzzy model design a fuzzy model controller to determine the wash time of domestic washing machine. Assume the inputs are dirt & grease on cloths. Use three descriptors for each input variable & five descriptors for output variables. Derive necessary membership functions & required fuzzy rules for the application	10	May 2025

Module 4: Introduction to Deep Learning

#	Question	Marks	Month & Year
#	Implement AND function using McCulloch-Pitts neuron (take binary data)	5	Dec 2023
#	Explain McCulloch-Pitts neuron with an example	5	Dec 2024
#	Explain the XOR function using McCulloch-Pitts model (use binary data representation)	10	May 2023
#	Implement AND function using McCulloch-Pitts neuron	5	May 2025
#	Explain the components of CNN architecture	10	Dec 2023
#	Explain Architecture of CNN in detail. List any two applications of it	10	Dec 2025
#	Briefly explain CNN architecture with detailed	10	May 2023
#	Describe the architecture of a typical CNN. What are the main components, and how do they contribute to the network's performance in image recognition tasks?	10	May 2024
#	Explain Architecture of CNN in detail. List any two applications of it	10	May 2025
#	Explain the components of ANN architecture	10	Dec 2024
#	Explain the component of ANN architecture	10	May 2025
#	Compare between ANN and RNN	5	Dec 2025
#	Compare between ANN and RNN	5	May 2025
#	Explain in detail the Long Short-Term Memory Network with an example	10	Dec 2024
#	What are all the objective of using LSTM? Explain its architecture	10	Dec 2025
#	Write short note on: LSTM	5	May 2023
#	State and elaborate the applications of deep learning	10	Dec 2023
#	What are the essential components of Autoencoder explain in detail? Which activation function is best for autoencoder	10	May 2023
#	Discuss different activation function used in Neural Network	5	May 2025

Module 5: Advanced ML Classification Techniques

#	Question	Marks	Month & Year
#	What is the significance of ROC curves?	5	Dec 2023
#	State Ensemble methods and describe anyone	10	Dec 2023
#	Explain Ensemble Methods	10	Dec 2024
#	How do you explain random forest? Does random forest need pruning, explain in detail?	10	May 2023
#	How Bagging and Boosting handle bias-variance trade-off differently, and analyze their effectiveness in dealing with noisy data and overfitting. Explain with algorithmic such as Random Forest and AdaBoost	10	Dec 2025
#	How Bagging and Boosting handle bias-variance trade-off differently, and analyze their effectiveness in dealing with noisy data and overfitting. Explain with algorithmic such as Random Forest and AdaBoost	10	May 2025
#	How does class imbalanced affect on classification? What are the ways to solve class imbalanced problem	10	May 2023
#	Explain bootstrap for sampling	5	Dec 2024
#	Write short note on: Holdout method	5	May 2023
#	Explain any 4 Metrics for evaluating classifier performance. Discuss on Hold out method and random sampling	10	Dec 2025
#	Explain any 4 Metrics for evaluating classifier performance. Discuss any two cross validation methods	10	May 2025
#	Which are the various metrics for evaluating the classifier performance?	10	May 2023
#	Explain Metrics for Evaluating Classifier Performance	10	Dec 2025
#	Calculate Accuracy, Precision, Recall, Sensitivity and Specificity for the following example (Actual Class Buys_Computer=yes/no vs Predicted Class Buys_Computer=yes/no: 6954/46, 412/2588)	10	Dec 2023
#	Define Accuracy, precision, and recall. Evaluate performance of classifier1 and classifier 2 on the basis of above evaluation parameters, given confusion matrix (Classifier 1: F/N/F 20/10/N 10/60; Classifier 2: F/N/F 0/15/N 5/80)	10	Dec 2024
#	Let's consider a binary classification problem where we have built a classifier to predict whether a transaction is fraudulent (positive class) or legitimate (negative class). After training the classifier and testing it on a dataset, we obtain the following confusion matrix: Actual Legitimate/Actual Fraudulent/Predicted Legitimate 850/30/Predicted Fraudulent 20/100. Calculate Accuracy, Precision, Recall and F1-score	10	May 2024

Module 6: Trends and Applications in Data Science

#	Question	Marks	Month & Year
#	Perform a case study on book recommendation system (data science based)	10	Dec 2023
#	Perform a case-study on video recommendation system (data science based)	10	Dec 2024
#	Write a short note on Data Science for Multi modal applications	10	Dec 2023
#	Explain Data science for Multimodal applications	10	May 2023
#	Briefly explain Data science for Multimodal applications	10	May 2024
#	Write short note on: Application of Data science for text	5	May 2023
#	Write a short note on Trends in Data Science for audio	10	Dec 2024
#	Explain the role of machine learning in multimodal applications. Discuss how different data modalities (text, audio, image, and video) are integrated to improve the performance and user experience of these systems with real time applications	10	Dec 2025
#	Explain the role of machine learning in multimodal applications. Discuss how different data modalities (text, audio, image, and video) are integrated to improve the performance and user experience of these systems with real time applications	10	May 2025